**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

## MARK SCHEME for the May/June 2014 series

# 4037 ADDITIONAL MATHEMATICS

**4037/12** Paper 1, maximum raw mark 80

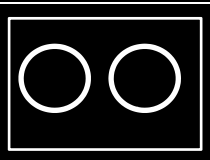
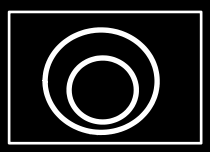
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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

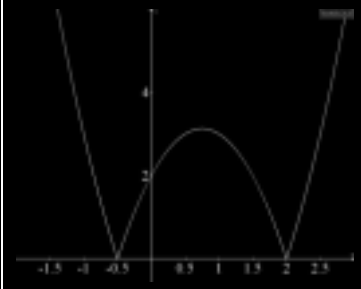
Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2014 series for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level components and some Ordinary Level components.

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| | | | |
|---|---|---|---|
| **1** | $\dfrac{\cos^2 A + (1+\sin A)^2}{(1+\sin A)\cos A}$ | **M1** | **M1** for obtaining a single fraction, correctly |
| | $\dfrac{\cos^2 A + 1 + 2\sin A + \sin^2 A}{(1+\sin A)\cos A}$ | **M1** | **M1** for expansion of $(1+\sin A)^2$ and use of identity |
| | $= \dfrac{2(1+\sin A)}{(1+\sin A)\cos A}$ | **DM1** | **DM1** for factorisation and cancelling of $(1+\sin A)$ factor |
| | $= \dfrac{2}{\cos A} = 2\sec A$ | **A1** | **A1** for use of $\dfrac{1}{\cos A} = \sec A$ and final answer |
| | **Alternative:** | | |
| | $\dfrac{\cos A(1-\sin A)}{(1+\sin A)(1-\sin A)} + \dfrac{1+\sin A}{\cos A}$ | **M1** | **M1** for multiplying first term by $\dfrac{1-\sin A}{1-\sin A}$ |
| | $= \dfrac{\cos A(1-\sin A)}{1-\sin^2 A} + \dfrac{1+\sin A}{\cos A}$ | **M1** | **M1** for expansion of $(1-\sin A)(1+\sin A)$ and use of identity |
| | $= \dfrac{\cos A(1-\sin A)}{\cos^2 A} + \dfrac{1+\sin A}{\cos A}$ | | |
| | $= \dfrac{1-\sin A}{\cos A} + \dfrac{1+\sin A}{\cos A}$ | **M1** | **M1** for simplification of the 2 terms |
| | $= \dfrac{2}{\cos A} = 2\sec A$ | **A1** | **A1** for use of $\dfrac{1}{\cos A} = \sec A$ and final answer |
| **2 (a) (i)** |  | **B1** | |
| **(i)** |  | **B1** | |
| **(b) (i)** | 6 | **B1** | |
| **(ii)** | 5 | **B1** | |
| **(iii)** | 9 | **B1** | |

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| 3 | (i) |  | **B1** | **B1** for shape |
| | | | **B1** | **B1** for $y = 2$ (must have a graph) |
| | | | **B1** | **B1** for $x = -0.5$ and 2 (must have a graph) |
| | (ii) | Maximum point occurs when $y = \frac{25}{8}$ | **M1** | **M1** for obtaining the value of $y$ at the maximum point, by either completing the square, differentiation, use of discriminant or symmetry. |
| | | so $k > \frac{25}{8}$ | **A1** | Must have the correct sign for **A1** Ignore any upper limits |
| 4 | | $\int_0^a \sin 3x \, dx = \frac{1}{3} \; dx = \frac{1}{3}$ | **B1,B1** | **B1** for $k \cos 3x$ **only**, **B1** for $-\frac{2}{3}\cos 3x$ **only** |
| | | $\left[-\frac{2}{3}\cos 3x\right]_0^a = \frac{1}{3}$ | **M1** | **M1** for correct substitution of the correct limits into their result |
| | | $\left(-\frac{2}{3}\cos 3a\right) - \left(-\frac{2}{3}\right) = \frac{1}{3}$ | **A1** | **A1** for correct equation |
| | | $\cos 3a = 0.5$ | **M1** | **M1** for correct method of solution of equation of the form $\cos ma = k$ |
| | | $3a = \frac{\pi}{3}$, $a = \frac{\pi}{9}$ | **A1** | **A1** allow 0.349, must be a radian answer |
| 5 | (i) | $2^{5x} \times 2^{2y} = 2^{-3}$ leads to $5x + 2y = -3$ | **B1, B1 DB1** | **B1** for $2^{2y}$, **B1** for $2^{-3}$, **B1** for dealing with indices correctly to obtain given answer |
| | (ii) | $7^x \times 49^{2y} = 1$ can be written as $x + 4y = 0$ | **B1** **B1** | **B1** for either $7^{4y}$ or $7^0$ seen **B1** for $x + 4y = 0$ |
| | | Solving $5x + 2y = -3$ and $x + 4y = 0$ leads to | **M1** | **M1** for solution of their simultaneous equations, must both be linear |
| | | $x = -\frac{2}{3}$, $y = \frac{1}{6}$ | **A1** | **A1** for both, allow equivalent fractions only |

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| 6 | (a) | **YX** and **ZY** | **B1,B1** | **B1** for each, must be in correct order, |
| | (b) | $\mathbf{B} = \mathbf{A}^{-1}\begin{pmatrix} 3 & 9 \\ -6 & -3 \end{pmatrix}$, | **M1** | **M1** for pre-multiplication by $\mathbf{A}^{-1}$ |
| | | $= -\frac{1}{3}\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} 3 & 9 \\ -6 & -3 \end{pmatrix}$ | **B1,B1** | **B1** for $-\frac{1}{3}$, **B1** for $\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$ |
| | | $= -\frac{1}{3}\begin{pmatrix} -9 & 3 \\ -18 & 21 \end{pmatrix}$ or $\begin{pmatrix} 3 & -1 \\ 6 & -7 \end{pmatrix}$ | **DM1**<br>**A1** | **DM1** for attempt at matrix multiplication<br>**A1** allow in either form |
| | | **Alternative method**: | | |
| | | $\begin{pmatrix} 5 & -2 \\ -4 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 3 & 9 \\ -6 & -3 \end{pmatrix}$ | **M1** | **M1** for a complete method to obtain 4 equations |
| | | Leads to $5a - 2c = 3$, $5b - 2d = 9$<br>$-4a + c = -6$, $-4b + d = -3$ | **A2,1,0** | –1 for each incorrect equation |
| | | Solutions give matrix | **M1** | **M1** for solution to find 4 unknowns |
| | | $-\frac{1}{3}\begin{pmatrix} -9 & 3 \\ -18 & 21 \end{pmatrix}$ or $\begin{pmatrix} 3 & -1 \\ 6 & -7 \end{pmatrix}$ | **A1** | **A1** for a correct, final matrix |

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| 7 | (i) | $\sin\frac{\theta}{2}=\frac{6}{8}$, $\frac{\theta}{2}=0.8481$ or better<br>or $12^2=8^2+8^2-128\cos\theta$ | **M1** | **M1** for a complete method to find either $\theta$ or $\frac{\theta}{2}$ |
| | | $\theta=1.6961$ or better | **A1** | Answer given. |
| | | or using areas<br>$\frac{1}{2}\times12\times2\sqrt{7}=\frac{1}{2}8^2\sin\theta$ oe<br>$\sin\theta=0.9922$, $\theta=1.4455$ or 1.6961 | **M1**<br>**A1** | **M1** for using the area of the triangle in 2 different forms<br>**A1** for choosing the correct angle. |
| | (ii) | Arc length = $(2\pi-1.696)\times8$ | **M1** | **M1** for correct attempt at a minor or major arc length |
| | | $(36.697$ or $36.7)$ | **A1** | **A1** for correct major arc length, allow unsimplified |
| | | Perimeter = $12+(2\pi-1.696)\times8$<br>$=48.7$ | **A1** | **A1** for 48.7 or better |
| | (iii) | Area = $\frac{8^2}{2}(2\pi-1.696)+\frac{8^2}{2}\sin1.696$ | **M1,M1** | **M1** for correct attempt to find area of major sector |
| | | $=178.5,\ 178.6,$ awrt 179 | **A1** | **M1** for correct attempt to find area of triangle, using any method |
| | | **Alternative:**<br>Area = $\pi8^2-\left(\frac{1}{2}8^2(1.696)-\frac{8^2}{2}\sin1.696\right)$ | | **M1** for attempt at area of circle – area of minor sector<br>**M1** for area of triangle |

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| **8 (a) (i)** | 720 | **B1** | |
| **(ii)** | 240 | **B1** | |
| **(iii)** | Starts with either a 2 or a 4: 48 ways | **B1** | allow unevaluated |
| | Does not start with either a 2 or a 4: 96 ways (i.e. starts with 1 or 5) | **B1** | allow unevaluated |
| | Total = 144 | **B1** | must be evaluated |
| | **Alternative 1:** | | |
| | Ends with a 2, starts with a 1,4 or 5 : 72 ways<br>Ends with a 4, starts with a 1,2 or 5 : 72 ways<br>Total =144 | **B1**<br>**B1**<br>**B1** | |
| | **Alternative 2**: | | |
| | $240-\left(2\times 2\times {}^{4}P_{3}\right)$ or $\left(4\times {}^{4}P_{3}\times 2\right)-\left(2{}^{4}P_{3}\right)$<br>$=144$ | **B2**<br>**B1** | **B2** for correct expression seen, allow *P* notation |
| | **Alternative 3:** | | |
| | ${}^{3}P_{1}\times {}^{4}P_{3}\times {}^{2}P_{1}$ or $3\times 4\times 2$<br>$=144$ | **B2**<br>**B1** | Allow *P* notation here, for **B2** |
| **(b)** | With twins : ${}^{16}C_{4}\ \left(=1820\right)$ | **B1** | |
| | Without twins: ${}^{16}C_{6}\ \left(=8008\right)$ | **B1** | |
| | Total: 9828 | **B1** | |
| | **Alternative:** | | |
| | ${}^{18}C_{6}-\left(2\times {}^{16}C_{5}\right)$<br>$=9828$ | **B1,B1**<br>**B1** | **B1** for ${}^{18}C_{6}-....,$ , **B1** for $2\times {}^{16}C_{5}$ |

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| **9** | **(i)** | $h = \dfrac{4000}{\pi r^2}$ or $\pi r^2 h = 4000$<br>$A = 2\pi rh + 2\pi r^2$ | **B1** | |
| | | $A = 2\pi r \dfrac{4000}{\pi r^2} + 2\pi r^2$ | **M1**<br>**A1** | **M1** for substitution of $h$ or $\pi rh$ into *their* equation for $A$<br>**A1** Answer given |
| | **(ii)** | $\dfrac{\mathrm{d}A}{\mathrm{d}r} = -\dfrac{8000}{r^2} + 4\pi r$ | **B1, B1** | **B1** for each term correct |
| | | When $\dfrac{\mathrm{d}A}{\mathrm{d}r} = 0$, $r^3 = \dfrac{8000}{4\pi}$ | **M1** | **M1** for equating to zero and attempt to find $r^3$ |
| | | leading to $A = 1395,\ 1390$ | **M1**<br>**A1** | **M1** for substitution of their $r$ to obtain $A$.<br>**A1** for 1390 or awrt 1395 |
| | | $\dfrac{\mathrm{d}^2 A}{\mathrm{d}r^2} = \dfrac{16000}{r^3} + 4\pi$,<br>which, is positive so a minimum. | √**B1** | √**B1** for a complete correct method and conclusion. |

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| **10 (i)** | Velocity $= 26 \times \frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$<br>$= 10\mathbf{i} + 24\mathbf{j}$ | **M1**<br>**A1** | **M1** for $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$ |
| | **Alternative 1**:<br>$\lvert 10\mathbf{i} + 24\mathbf{j}\rvert = \sqrt{10^2 + 24^2}$<br>$= 26$ | **M1** | **M1** for working from given answer to obtain the given speed |
| | Showing that one vector is a multiple of the other, hence same direction | **A1** | **A1** for a completely correct method |
| | **Alternative 2**:<br>$\sqrt{5^2 + 12^2} = 13$, $13k = 26$, so $k = 2$<br>Velocity $= 2(5\mathbf{i} + 12\mathbf{j})$, | **M1** | **M1** for attempt to obtain the 'multiple' and apply to the direction vector |
| | Velocity $= 10\mathbf{i} + 24\mathbf{j}$ | **A1** | **A1** for a completely correct method |
| | **Alternative 3**:<br>Use of trig: $\tan\alpha = \frac{12}{5}$, $\alpha = 67.4°$<br>Velocity $26\cos 67.4°\mathbf{i} + 26\sin 67.4\mathbf{j}$ | **M1** | **M1** for reaching this stage |
| | Velocity $= 10\mathbf{i} + 24\mathbf{j}$ | **A1** | **A1** for a completely correct method |
| **(ii)** | Position vector $= 4(10\mathbf{i} + 24\mathbf{j})$<br>or $40\mathbf{i} + 96\mathbf{j}$ | **B1** | Allow either form for **B1** |
| **(iii)** | $(40\mathbf{i} + 96\mathbf{j}) + (10\mathbf{i} + 24\mathbf{j})t$ oe | **M1**<br>**A1** | **M1** for *their* $(\mathbf{ii}) + (10\mathbf{i} + 24\mathbf{j})t$ or $(10\mathbf{i} + 24\mathbf{j}) \times (t + 4)$<br>**A1** correct answer only |
| **(iv)** | $(120\mathbf{i} + 81\mathbf{j}) + (-22\mathbf{i} + 30\mathbf{j})t$ oe | **B1** | |
| **(v)** | $40 + 10t = 120 - 22t$ or<br>$96 + 24t = 81 + 30t$ | **M1** | **M1** for equating like vectors |
| | $t = 2.5$ or 18:30 | **A1** | **A1** Allow for $t = 2.5$ |
| | Position vector $= 65\mathbf{i} + 156\mathbf{j}$ | **DM1**<br>**A1** | **DM1** for use of $t$ to obtain position vector<br>**A1** cao |

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| **11 (a)** | $\tan x(\tan x + 5) = 0$<br>$\tan x = 0,\quad x = 0°, 180°$<br>$\tan x = -5,\quad x = 101.3°$ | **B1,B1**<br>**B1** | **B1** for each , must be from correct work |
| **(b)** | $2(1-\sin^2 y) - \sin y - 1 = 0$<br>$2\sin^2 y + \sin y - 1 = 0$<br>$(2\sin y - 1)(\sin y + 1) = 0$ | **M1** | **M1** for use of correct identity and attempt to solve resulting 3 term quadratic equation. |
| | $\sin y = \frac{1}{2}$, $y = 30°, 150°$ | **A1,A1** | |
| | $\sin y = -1$, $y = 270°$ | **A1** | |
| **(c)** | $\cos\left(2z - \frac{\pi}{6}\right) = \frac{1}{2}$ | **M1** | **M1** for dealing with sec correctly and obtaining $\frac{\pi}{3}$ or 1.05 |
| | $\left(2z - \frac{\pi}{6}\right) = \frac{\pi}{3}$<br>$z = \frac{\pi}{4}$ or 0.785 or better | **A1** | |
| | $\left(2z - \frac{\pi}{6}\right) = \frac{5\pi}{3}$ | **M1** | **M1** for obtaining a second equation $\left(2z - \frac{\pi}{6}\right) = 2\pi - \textit{their}\ \frac{\pi}{3}$ oe |
| | $z = \frac{11\pi}{12}$ or 2.88 or better | **A1** | |

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